

Airbus Amber

ED-247 (Rev. B) for Training

RAeS Flight Simulation Conference 2024

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22-23rd October 2024

AIRBUS

ED-247 rev. B
Agenda

- ❑ A standard for what?
- ❑ ED-247 partners
- ❑ ED-247 Overview
- ❑ ED-247 Status
- ❑ ED-247 Wish list
- ❑ ED-247 rev. B: Deep Dive for training purpose
- ❑ ED-247 rev. B: Practical Things

ED-247 : A standard for what?

To provide a framework that supports the demonstration of the interoperability of virtual avionic components.

To define a global process that describes the exchange, integration and exploitation of virtual avionics components and their environments.

*The task of the EUROCAE WG-97 working group is to propose a **win-win strategy** for integrator and avionics suppliers by **setting a standard** to **increase the reuse** of virtual equipment, enabling test bench **plug-and-play integration**.*

ED-247 - Partners

Integrators

Systems

Architecture

AIRBUS

THALES

United Electronic Industries

DASSAULT AVIATION

Collins Aerospace

rti

AIRBUS HELICOPTERS

FlightSafety international

dSPACE

AIRBUS DEFENCE & SPACE

techSAT mastering integration complexity

Core Team

BOEING

BNAE

NATIONAL INSTRUMENTS

COMAC

DIEHL Aerospace

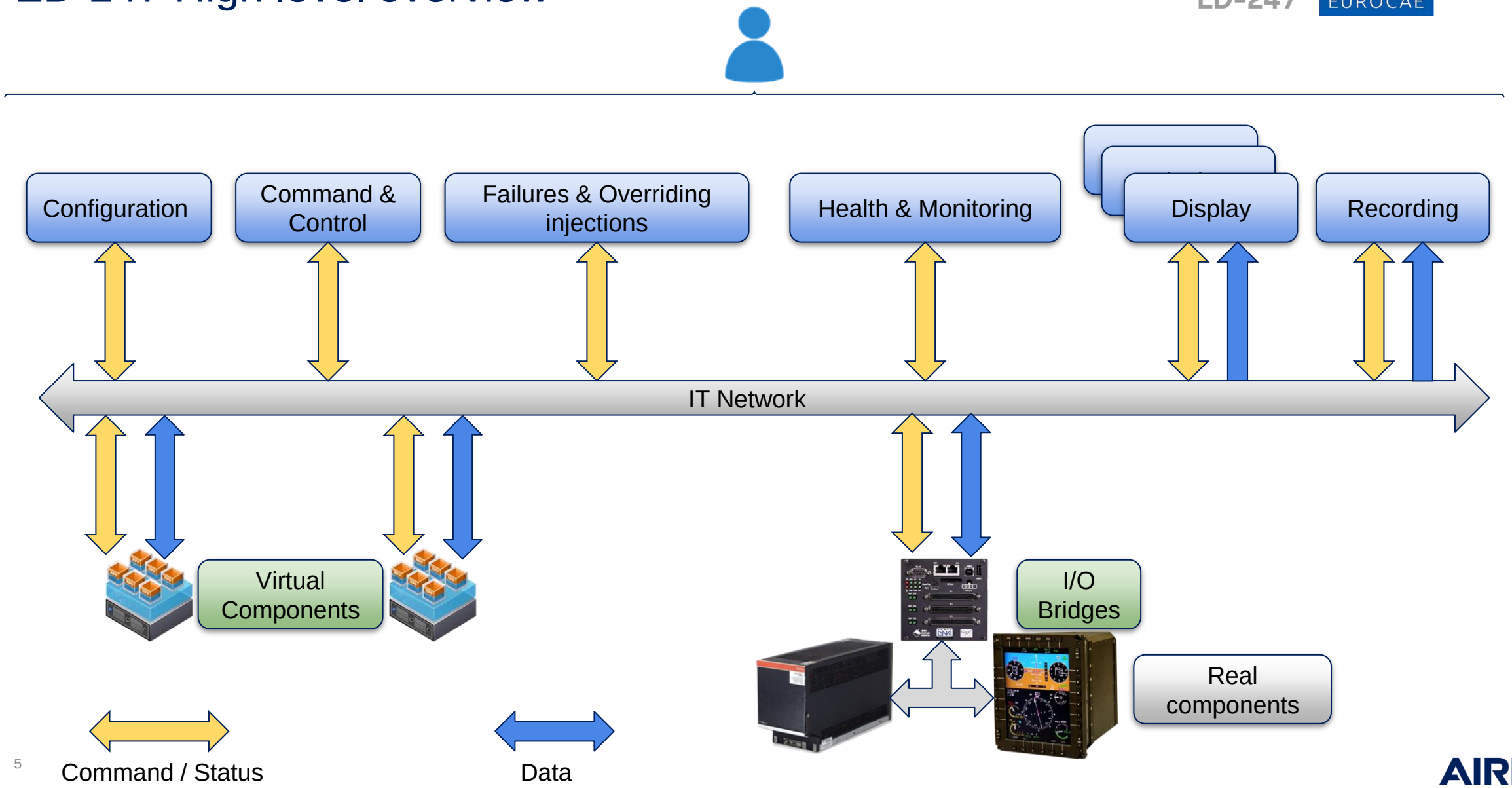
SCALIAN

ScioTeq

VOLANT AEROTECH | 沃兰特

EASA

ED-247 High level overview



ED-247 Status

2013
- ❑ Start

Nov. 2017
- ❑ ED-247 1st Release

Scope
Virtualization of usual avionics protocols

Topology
- Ethernet / UDP / MCAST

Signals
- AFDX
- A429
- CAN
- DISCRETE
- ANALOG
- NAD (Non Avionics Data)

April 2020
- ❑ ED-247 Revision A publication

Scope
- 1st Release
- **Hybrid benches**
- **Communication Model**
- **Resolution process**

Topology
- 1st Release
- **UCAST**

Signals
- 1st Release
- **Serial**
- **Voice (packets)**
- **Video (packets)**
- **MIL-STD-1553**

Dec 2023
- ❑ ED-247 Revision B publication

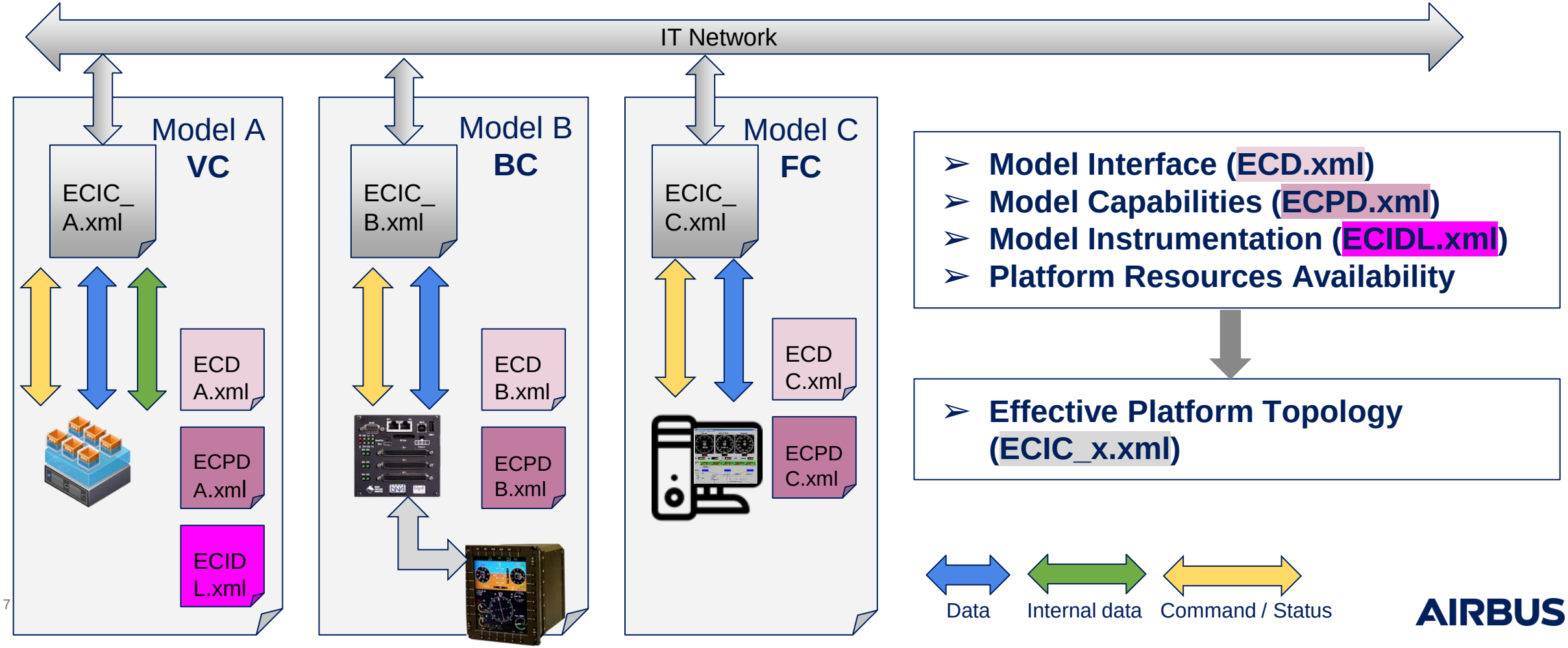
Scope
- Rev. A
- **C&C**
- **Time Mgt.**
- **State Machine**
- **Health & Monitoring**
- **Instrumentation**
- **Component Delivery**

Topology
- Rev. A
- DDS
- **TCP**

Signals
- Rev. A

ED-247 rev. B: training perspectives - Resolution Process

Connections are at bus level => Bus contents - labels definitions - **are ICD related**



ED-247 rev. B: training perspectives
Data Frame Format

ED-247 Transport Header

ED-247 Data Frame Header

ED-247 Data Frame Payload

Specific features not supported by the transport mean (e.g. Segmentation)

Frame Embedded specific ED-247 features + Extensions for future revisions

- Component Identifier (ECIID)
- QoS (Sequence Number & Transport Time Stamp (Time of the emission of the frame))
- Overridden information (if flag set in Data Frame Header)
- Future extensions defined in the Data Frame Header

Original Avionics format (e.g. AFDX) + ED-247 Features for platform reliability

Offset Byte	Byte Bits	0								1								2								3							
		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
0	0	0xED02																0x14								1	0	0	0	0	0	0	0
4	32	EC INSTANCE IDENTIFIER (ECIID)																SEQUENCE NUMBER (SN)															
8	64	TRANSPORT TIMESTAMP (TTS)																															
12	96																																
16	128	ECIID																FOI ID															
20	160	ED-247 Data frame payload																															

ED-247 rev. B: training perspectives
Payload

Focus on functional data Payload : Original Avionics + Precise Timing Information

Offset Byte	Byte Bits	0								1								2								3							
		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
0	0	TRANSPORT STREAM x UID																															
4	32	TRANSPORT STREAM x PAYLOAD SIZE																															
8	64	DATA TIMESTAMP (DTS)																															
12	96																																
16	128	SAMPLE 0 DATA (data of length x0)																															
16+x0/8	128+x0	SAMPLE 1 DATA TIME OFFSET (in nanoseconds)																															
20+x0/8	160+x0	SAMPLE 1 DATA (data of length x1)																															
20+(x0+x1)/8	160+x0+x1	...																															
0	0	TRANSPORT STREAM y UID																															
4	32	TRANSPORT STREAM y PAYLOAD SIZE																															
8	64	DATA TIMESTAMP (DTS)																															
12	96																																
16	128	SAMPLE 0 DATA (data of length y0)																															
16+y0/8	128+y0	SAMPLE 1 DATA TIME OFFSET (in nanoseconds)																															
20+ y0/8	160+x1	SAMPLE 1 DATA (data of length y1)																															
20+(y0+y1)/8	160+y0+y1	...																															
...	...	...																															

ED-247 mandatory field ED-247 optional fields

The Payload is made of several (one or more) consecutive samples with the time of emission of each sample

Samples are segregated according to their source with the field Transport Stream UID

ED-247 rev. B: training perspectives
Architecture optimizations for resources adaptations

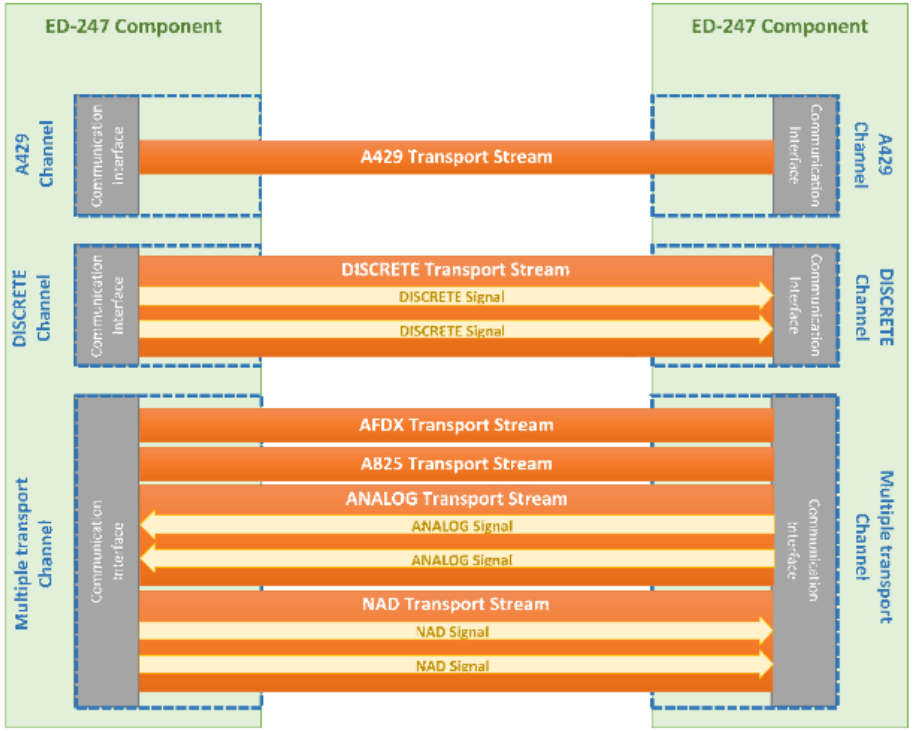
Topology: an EC can declare which streams are groupable => Bandwidth and Interruptions optimization without loss of fidelity

Resources: Data , command & notification, and instrumentation are on separated busses

Options policy: ED-247 defines capabilities in several domains :

- ➢ Communication interface (TCP - UDP - DDS)
- ➢ Emission periods
- ➢ Packetization strategies
- ➢ Computational Frequency Groups

…



➡ Best suited virtual components for training needs

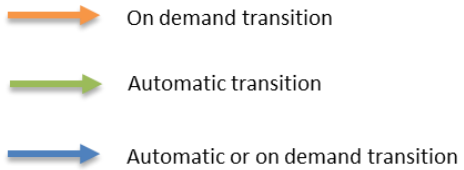
ED-247 rev. B: training perspectives
State Machine - Command protocol optimized to reduce latency

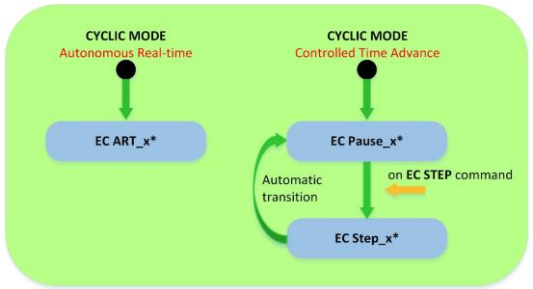
The State-Machine is designed for distributed architectures

Real time modes
- Accurate
- Accelerated
- Decelerated

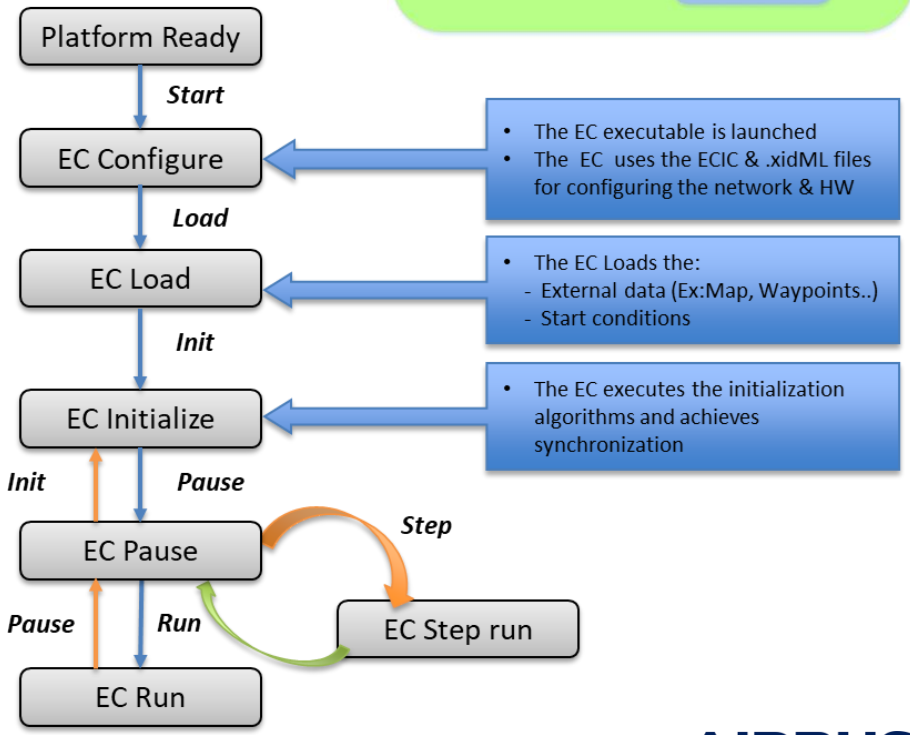
Non Real time modes
- Step-by-steps
- As Fast As Possible
- Continuous-Controlled
- Just in Time

- Non real time modes are optional
- A device controller pilots the ECs
- An autonomous and distributed sequencing mean is mandatory





Simplified version

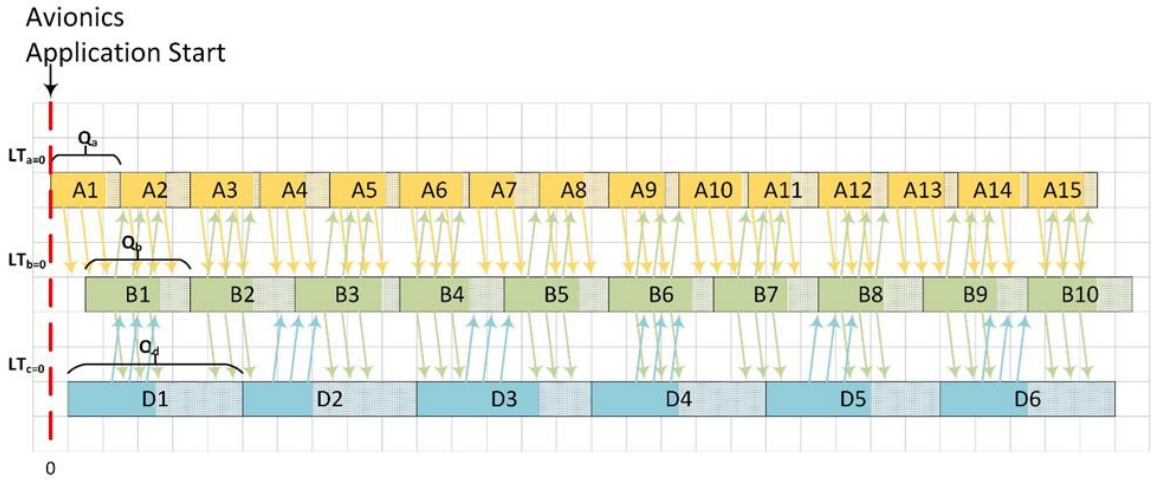


ED-247 rev. B: training perspectives
Time Management

It corresponds to the implementation of the modes offered by the ED-247 state machine

It is designed to cover aircraft avionics architectures :

- Asynchronous
- Periodic
- I/O and applications processing segregated
- Drift and jitter

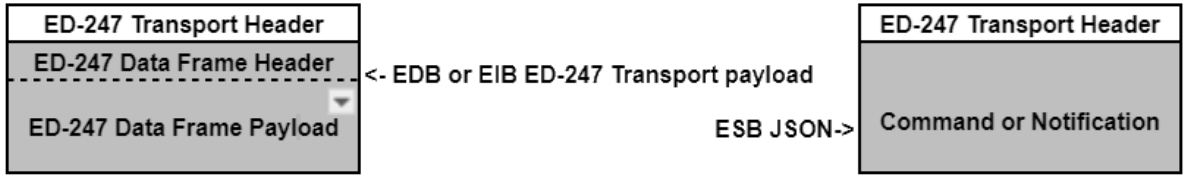


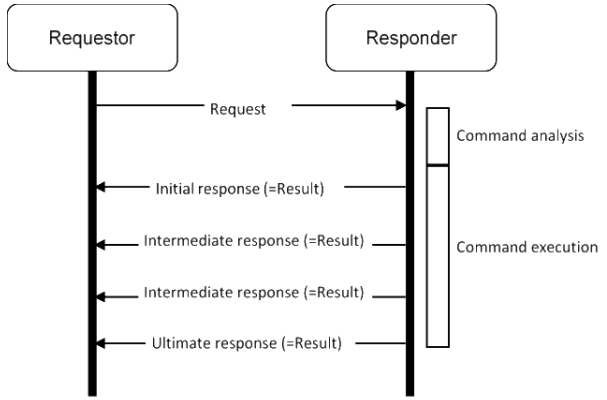
The standard provides all definitions needed to pilot ECs in all state machine modes

Fully event based components shall be approximated through high frequency calls

ED-247 rev. B: training perspectives
Commands & Notifications

Principle : One generic Command-and-Control mechanism/Protocol everywhere the standard needs it



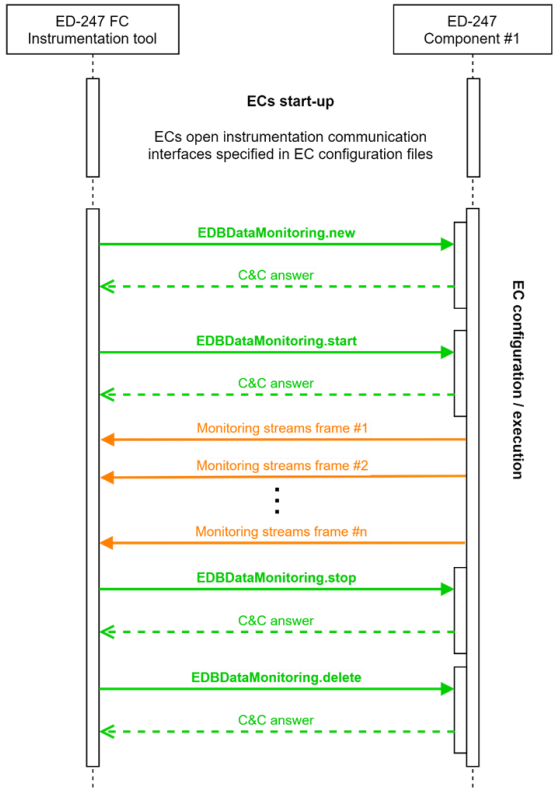


- An ED-247 command is a request waiting for one or more responses, allowing immediate and one or more steps answers
- Based on JSON-RPC 2.0 with added features to cover all ED-247 definitions
- TCP , UDP and DDS are supported
- Errors classification (opened) for all commands
- Strict grammar based on JSON Objects
- Notifications have the same format as command

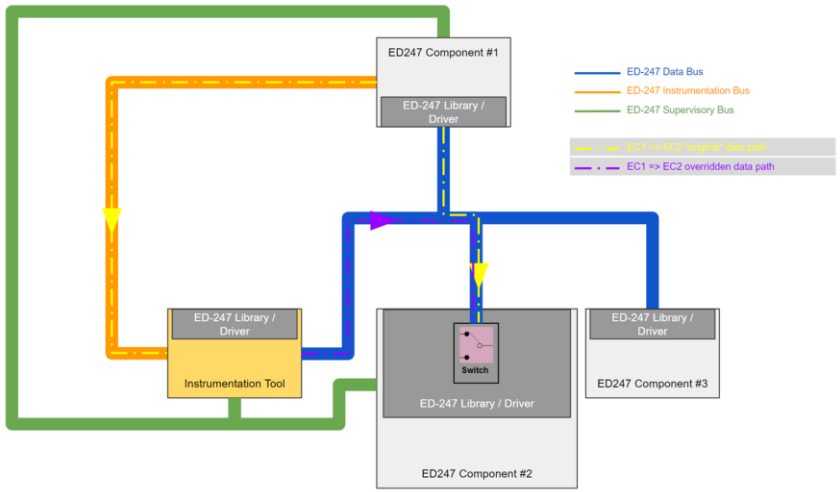
```
{
    "BlockType": "Command",
    "StandardRevision": "B",
    "RequestorECIID": 120,
    "CommandID": 2,
    "Method": "MyCommand",
    "RespondersECIID": [23,65,187],
    "Params": {
        "StreamID":[
            42,
            67,
            120
        ],
        "Timestamp": true
    }
}
```

ED-247 rev. B: training perspectives
Instrumentation

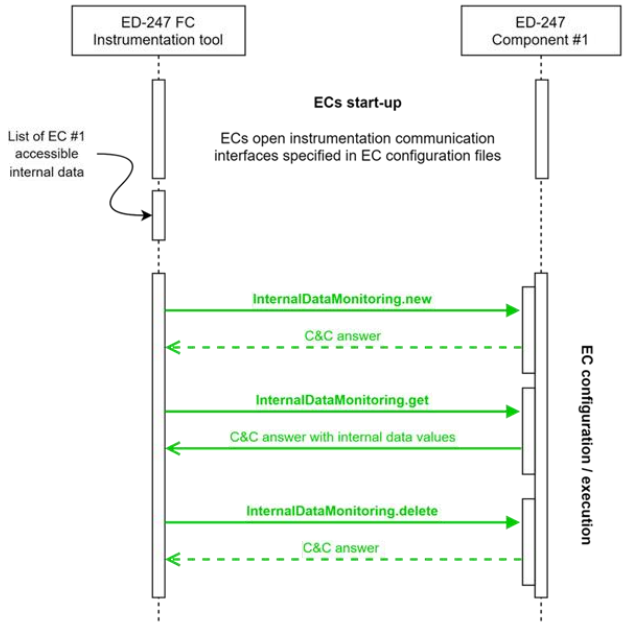
❑ *Functional data monitoring*



❑ *Failures injections and data override*



❑ *Internal data monitoring*



ATLAS : ED-247 is the official standard used by AIRBUS for training products

ED-247 rev. B training perspectives

Practical things

- The format for component delivery is defined by a .zip file with a predefined structure
- The standard defines an xml grammar for ED-247 integration files: ***Xsd (xml schema) are provided***
- ED-247 options supported shall be decided at contractual level
- ED-247 rev. A & rev. B frames are not compatible

- **Operational deployment:**
  - Open-source Airbus Commercial A/C ED-247 implementation: https://github.com/airbus/ED247_LIBRARY
  - wireshark ED-247 Dissector

Protocols covered in rev. B:

- **A429**
- **A825**
- **MIL-STD-1553B**
- **A453 / A708**
- **Serial Lines**
- **Audio Lines**
- **Video Lines**
- **Ethernet**
- **Discrete**
- **Analog**
- **Non Avionics data**

Thank you

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